

State-Conditioned Gate Arbitration in SAT/CDCL

Execution of the Preregistered MAAT v1.8 Gate Challenge Cycle II

Christof Krieg

MAAT Project, Independent Researcher
maat-research.com

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Abstract

Paper 75 preregistered a state-conditioned SAT/CDCL arbitration protocol before external execution and archived it at Zenodo (DOI: [10.5281/zenodo.21767033](https://doi.org/10.5281/zenodo.21767033)). This paper executes that frozen protocol on 98 hashed SATLIB instances: 19 calibration instances and 79 held-out test instances across six families. No support equation, threshold, policy coefficient, split, budget, null model, bootstrap rule, or status criterion was changed after preregistration.

The construction passes the frozen distinctness precondition and is not reconstructed by the declared classical summary model. Its realised family-instance-weighted test activation is 0.12558, narrowly above the minimum $q_{\text{test,min}} = 0.125$. Operationally, however, the result is negative. Against MOMS, the primary state-conditioned policy has family-balanced total-cost difference $\Delta U = -642.96$ with 95% bootstrap interval $[-841.63, -295.82]$, where positive values favour the state-conditioned policy. Relative regret against MOMS is 0.9135 (95% interval $[0.6956, 1.1756]$), satisfying the preregistered definition of **harmful**. The primary policy also loses significantly to the static v1.7 gate and score-with- R baselines. It beats the shuffled-gate null but not the matched-random-activation null. The frozen status tuple is therefore **executed / distinct / adequate_activation / harmful / negative_result**. A secondary decomposition shows that structural overhead amplifies the failure, but search cost alone also loses to MOMS. This is a negative external execution result, not a retuned gate proposal.

1 Purpose and Release Barrier

The question is whether structural supports evaluated on the evolving CDCL state can arbitrate usefully between a structural Mode-A brancher and a classical MOMS fallback. Paper 75 fixed the complete experiment before any external state-conditioned run. The exact release was published on 3 August 2026 as version 1.5.3 and assigned DOI [10.5281/zenodo.21767033](https://doi.org/10.5281/zenodo.21767033). Paper 76 began only after that public timestamp.

Three immutable identifiers guard the execution:

Paper-75 DOI = 10.5281/zenodo.21767033,
protocol SHA256 = 9e86b1e1f8ddcd3228c90f62cc9465d3...,
manifest SHA256 = 941fad18c5ea4839ee3aac4b5d7f1e3d....

The external gatekeeper also required all 98 raw-file hashes, DIMACS parseability, and absence of unknown CNFs to pass. All checks passed before the solver was called.

2 Frozen Protocol

This section summarises the executed contract; the preregistration remains the authoritative specification [1].

2.1 Decision state and supports

Supports are evaluated after unit propagation reaches a conflict-free fixpoint and immediately before each branch decision. The state contains original and persistent learned clauses, the partial assignment, decision levels, trail, reason clauses, VSIDS activity, and saved phase. Satisfied clauses are removed and falsified literals are deleted to form the residual CNF F_t .

On active residual variables U_t , the frozen supports are

$$H[X_t] = \left(1 + \text{mean}_{v \in U_t} \frac{|p_v - n_v|}{p_v + n_v + 10^{-9}}\right)^{-1}, \quad (1)$$

$$B[X_t] = \left(1 + \frac{\text{std}_{v \in U_t}(d_v)}{\text{mean}_{v \in U_t}(d_v) + 10^{-9}}\right)^{-1}, \quad (2)$$

$$S[X_t] = \frac{\text{mean}_{v \in U_t}(s_v)}{\max_{v \in U_t} s_v + 10^{-9}}, \quad (3)$$

$$V[X_t] = \text{clip}_{[0,1]}(0.65f_{\text{LCC}} + 0.35\bar{c}). \quad (4)$$

Here p_v, n_v are residual positive and negative occurrence counts, $d_v = p_v + n_v$, s_v is inverse-length-weighted pressure from residual clauses of length at most three, f_{LCC} is the largest component fraction, and $\bar{c}$ is mean local clustering. The emergent closure is

$$R_{\text{rob}} = \min\left((HBV)^{1/3}, (HBSV)^{1/4}\right). \quad (5)$$

2.2 Calibration and arbitration

Only MOMS trajectories from the 19 calibration instances determine weighted medians, MADs, the gate threshold, tie fraction, distinctness, and classical reconstruction. At most 128 deterministically selected states are retained per instance; 667 states were retained in this execution. The primary gate is

$$g_t = z_R + 0.50z_t, \quad G_{\text{gate}} = (1 + e^{-g_t})^{-1}. \quad (6)$$

The weighted lower 0.75 quantile produced

$$\theta_q = 0.8816963, \quad f_{\text{tie}} = 0.7287115. \quad (7)$$

The calibration median and MAD were 0.7627958 and 0.0446364 for R_{rob} , and -0.0018129 and 0.0089476 for ΔR_{rob} .

The gate routes to Paper-63 Mode A when active and to MOMS otherwise. Mode A uses the frozen candidate score

$$A_t(v) = P_t(v) - 0.38h_{\text{MAAT},t}(v) \quad (8)$$

with softmax inverse temperature 8. Directly compared stochastic policies use the same state-bound random number whenever they reach the same state.

2.3 Causal updates and overhead

The implementation maintains residual literal counts, short-clause pressure, and graph-edge multiplicities incrementally. Enqueue operations update the counters; assignment effects are reversed on backtrack; learned clauses remain persistent. Every 100th decision compares incremental supports against full reconstruction. The maximum discrepancy observed in the complete run was 7.27×10^{-14} , below the frozen 10^{-12} abort tolerance.

Each policy-instance pair has 5000 conflicts and 30 seconds. Search cost is

$$C_{\text{search}} = N_{\text{conflict}} + 0.30N_{\text{decision}} + 0.004N_{\text{propagation}} + 5000\mathbf{1}[\text{timeout}]. \quad (9)$$

Calibration fixed $\kappa_{\text{cal}} = 268.943$ search-cost units per second. For structural policies,

$$C_{\text{total}} = C_{\text{search}} + \kappa_{\text{cal}}T_{\text{structural}}. \quad (10)$$

Classical baselines report tracing timers separately but are not charged for diagnostic work that their decision rules do not require.

3 Data and Gatekeeper

The manifest contains AIM (20), Dubois (13), pigeonhole (5), flat graph colouring (20), satisfiable random 3-SAT UF20 (20), and unsatisfiable random 3-SAT UUF50 (20), sourced from SATLIB [2]. Frozen family-local hash ordering assigns 19 instances to calibration and 79 to testing.

Table 1: Pre-execution gatekeeper.

Check	Status
Published Paper-75 release and DOI	PASS
Protocol and manifest SHA256	PASS
98 expected CNFs present	PASS
98 instance SHA256 values	PASS
DIMACS parseability	PASS
No unexpected CNFs	PASS

Raw SATLIB CNFs are not redistributed. Only source metadata, hashes, and derived solver measurements are included with the experiment.

4 Construct and Activation Status

No support pair approached the frozen degeneracy boundary $|\rho_S| > 0.99$. The largest absolute weighted Spearman correlation was 0.6821 between B and S ; weighted support standard deviations ranged from 0.0769 to 0.1507. The construct therefore passed distinctness.

The preregistered 18-predictor classical reconstruction produced negative held-out weighted R^2 for all targets, including R_{rob} and G_{gate} . These very negative values indicate unstable extrapolation by the declared OLS model, not “negative predictability.” They are nevertheless below the frozen convergent-rediscovery threshold 0.90, so `construct_status=distinct`.

Family-instance-weighted activation was 0.1255808. This narrowly exceeds the predeclared minimum $q_{\text{test,min}} = 0.125$ while remaining far below the calibration target $q = 0.25$. Accordingly, `activation_status=adequate_activation`; the proximity to the boundary is reported without threshold adjustment.

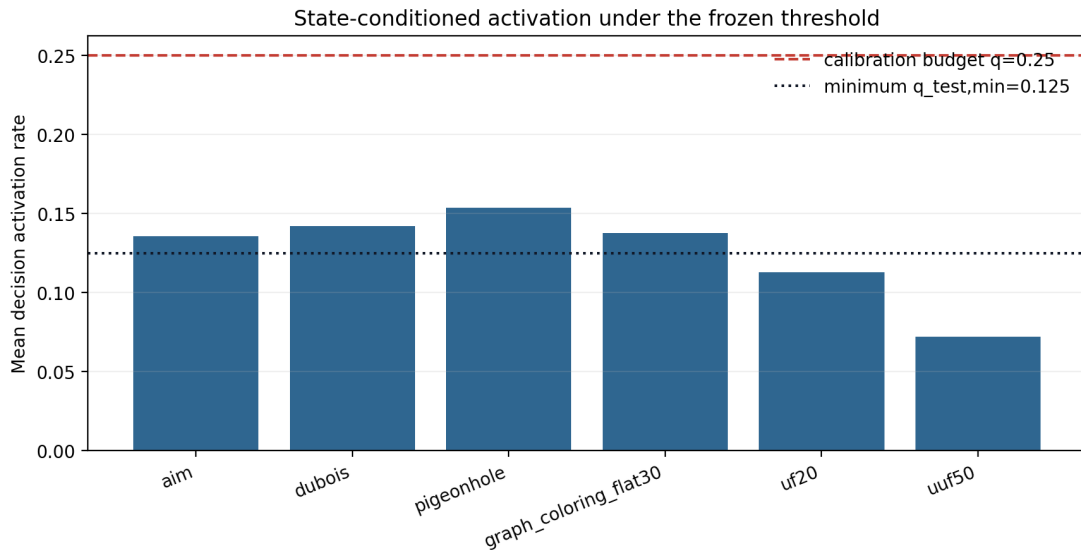


Figure 1: Realised primary activation by family. The frozen threshold is not refitted on test states.

5 Primary Utility Results

For policy A against comparator B , the registered direction is

$$\Delta U_i(A, B) = C_B(i) - C_A(i), \quad (11)$$

so positive values favour the state-conditioned policy. Confidence intervals use 10,000 paired instance-level bootstrap resamples within family followed by equal family aggregation.

Table 2: Primary state-v1.8 comparisons. Wins and losses refer to the primary policy; means and intervals are family-balanced.

Comparator	Mean ΔU	95% CI	Wins	Losses
Static v1.7 gate	-517.20	[-704.89, -177.72]	54	25
Score-with- R	-522.90	[-715.03, -172.41]	53	26
Shuffled gate	42.24	[6.65, 70.14]	43	36
Matched random activation	23.96	[-8.27, 51.14]	46	33
MOMS	-642.96	[-841.63, -295.82]	5	74
Jeroslow-Wang	-533.75	[-727.05, -209.60]	35	44
VSIDS	-862.58	[-1605.14, -274.01]	47	32

The apparent contrast between majority wins against the two prior MAAT baselines and strongly negative family-balanced means is a tail effect: fewer, costly losses dominate the average. Against MOMS the primary policy wins only five of 79 paired test cases.

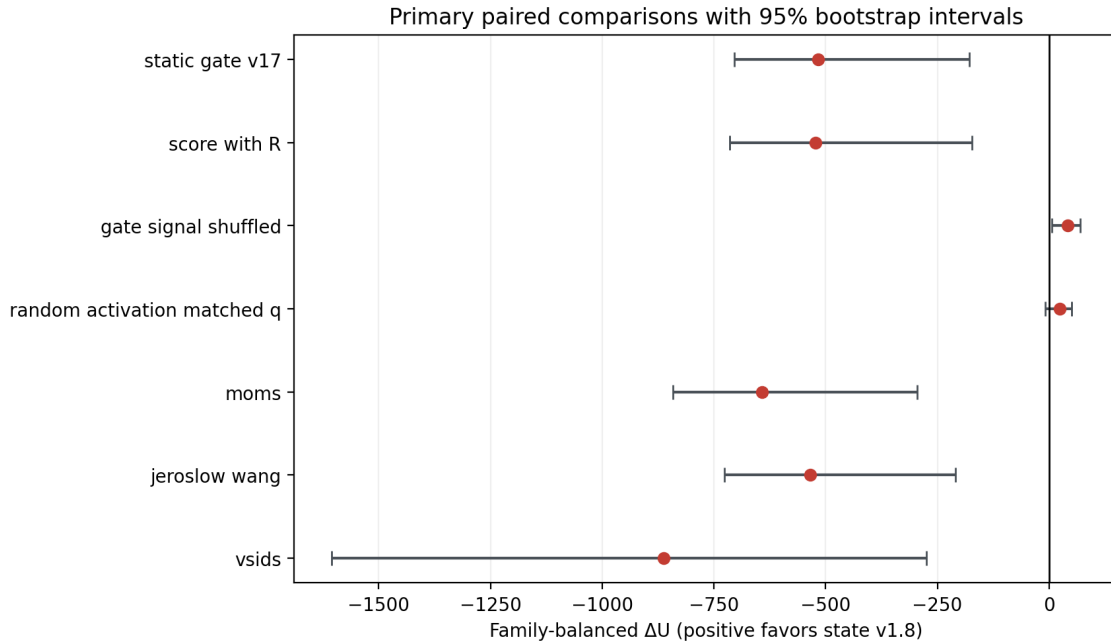


Figure 2: Primary paired comparisons. Positive values favour state v1.8.

The primary policy significantly beats the shuffled-gate null, showing that the realised gate sequence is not equivalent to the declared family/split shuffle. It does not significantly beat matched random activation. The minimum-support rule requires positive lower confidence bounds against both prior MAAT baselines and both nulls; it is therefore not met.

6 Safety and Cost Decomposition

Relative regret against the frozen MOMS fallback is

$$r_i = \frac{C_{\text{state}}(i) - C_{\text{MOMS}}(i)}{\max(C_{\text{MOMS}}(i), 10^{-9})}. \quad (12)$$

The family-balanced mean is 0.91349 with interval [0.69560, 1.17557]. Because the lower endpoint exceeds the frozen no-harm bound $\delta = 0.05$, the registered safety label is **harmful**. This is positive evidence of harm under the registered cost model, not merely failure to certify no harm.

A secondary, non-reclassifying decomposition separates search decisions from charged structural overhead. Against MOMS, search cost alone yields $\Delta U = -49.42$ with interval $[-69.81, -30.82]$. Total cost yields -642.96 with interval $[-841.63, -295.82]$. Mean measured structural overhead for the primary policy is 0.7054 seconds per test instance. Thus overhead is the dominant damage channel, especially on timeout trajectories, but removing it would not reverse the negative search result.

Table 3: Descriptive family means versus MOMS. Positive deltas favour state v1.8; this table does not alter the preregistered classification.

Family	Search ΔU	Total ΔU	Overhead penalty
AIM	0.67	-7.41	8.08
Dubois	-275.44	-348.66	73.22
Pigeonhole	-18.51	-3487.42	3468.91
Flat30	0.00	-1.40	1.40
UF20	-0.11	-0.57	0.46
UUF50	-3.10	-12.31	9.21

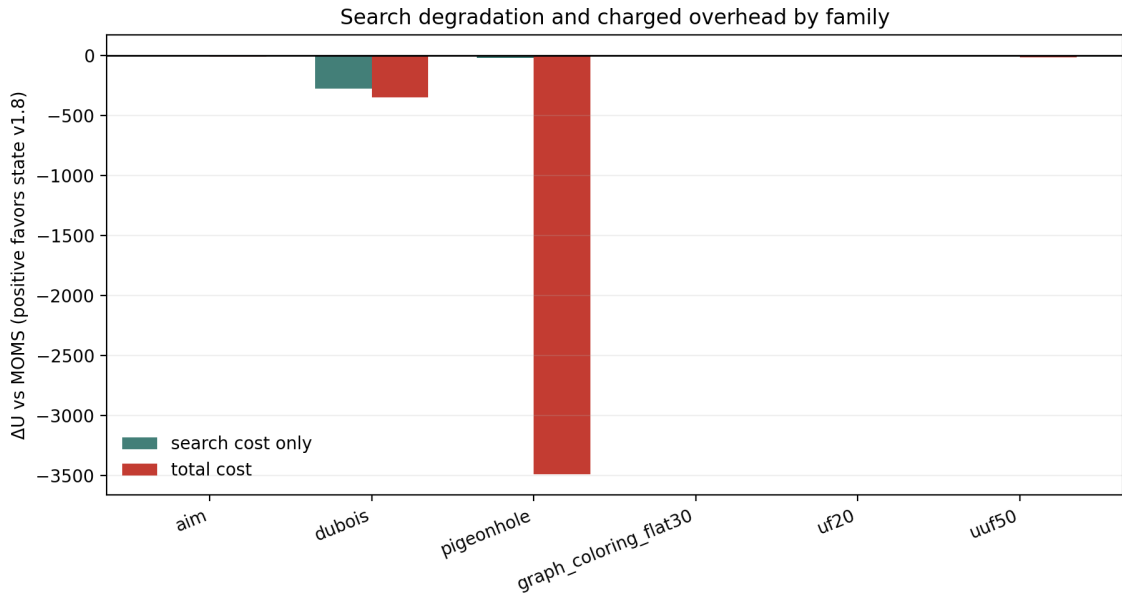


Figure 3: Search-only and overhead-inclusive family deltas versus MOMS. The pigeonhole tail dominates charged overhead, while Dubois also exhibits genuine search degradation.

7 Complete Frozen Outcome

Paper 75 requires independent status axes rather than a single favourable or unfavourable label.

Table 4: Paper-76 outcome under the frozen status system.

Axis	Frozen result
Execution	<code>executed</code>
Construct	<code>distinct</code>
Activation	<code>adequate_activation</code>
Safety	<code>harmful</code>
Utility	<code>negative_result</code>

The negative utility label follows independently from the confidence intervals against static v1.7 and score-with- R . The harmful safety label follows from relative regret against MOMS. Passing distinctness and activation does not override either result.

8 Interpretation

The experiment rejects the practical utility of this frozen state-conditioned arbitration on the declared benchmark. It does not imply that every dynamic SAT state descriptor is useless, nor does it test an industrial solver. It does show that three appealing properties—nondegenerate supports, a nontrivial gate, and partial superiority to a shuffled null—are insufficient for decision utility.

Two failures coexist. First, Mode-A activation degrades search in important families, most clearly Dubois. Second, continuously maintaining residual graph and support information is expensive on long timeout trajectories. A fallback architecture cannot claim safety merely because it activates infrequently: here activation barely clears the registered minimum yet still produces a harmful relative-regret interval.

No threshold, support, data split, budget, or status rule is modified in this paper. Any future attempt to reduce graph-update overhead or alter the active policy would constitute a new hypothesis and require a new preregistration.

9 Limitations

- The transparent Python CDCL backend is auditable but not an industrial solver such as CaDiCaL or Kissat. Results concern policy comparison within this shared backend.
- The benchmark contains six public SATLIB families but only 98 instances. The pigeonhole family has four test cases and therefore yields uncertain tail estimates despite equal-family aggregation.
- The OLS reconstruction model extrapolates poorly. Its negative R^2 values satisfy the frozen non-reconstruction condition but do not establish a strong positive notion of structural novelty.
- Runtime depends on the execution environment. The primary registered cost converts measured overhead using calibration-derived κ_{cal} ; raw wall time and operation-count search cost are both retained.
- Secondary ablations are descriptive only. None may replace the failed primary policy or promote its status.

10 Data, Licensing, and Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Experiment folder:

`experiments/maat_v18_state_conditioned_execution_paper76/`

The release includes code, the frozen metadata manifest, validation JSON, derived CSV/JSON outputs, figures, tests, and this manuscript. Raw SATLIB CNFs and source archives are intentionally excluded. Users must obtain them from SATLIB and verify them against the included SHA256 manifest. SATLIB authors and maintainers do not endorse MAAT.

Paper 76, its figures, and the derived result tables are released under the Creative Commons Attribution 4.0 International license (CC BY 4.0). Repository source code remains under the MIT License. Paper 75 is likewise archived under CC BY 4.0 on Zenodo. External SATLIB benchmark materials are not included and retain their original terms. The manifest provides attribution, source URLs, and integrity metadata but grants no additional rights in those external instances. The derived output tables contain solver measurements and metadata, not clause lists.

Reproduction begins with

```
python3 paper76_execution.py --data-dir /path/to/data_external --validate-only
```

and proceeds only after VALIDATION PASS. The complete run is

```
python3 paper76_execution.py --data-dir /path/to/data_external  
python3 paper76_report.py
```

Synthetic unit tests are executed with

```
PYTHONDONTWRITEBYTECODE=1 python3 -m unittest discover -s tests -v
```

11 Conclusion

Paper 76 completes the first execution of MAAT v1.8 Gate Challenge Cycle II. The experiment passed release, data, implementation, distinctness, and minimum activation checks. It nevertheless produced a harmful safety result and a negative utility result. The frozen state-conditioned gate is therefore not supported as a practical SAT/CDCL improvement under this protocol.

This outcome is accepted without modifying the preregistered hypothesis.

References

- [1] C. Krieg, “State-Conditioned Gate Arbitration in SAT/CDCL: A Preregistration for MAAT v1.8 Gate Challenge Cycle II,” Paper 75 (2026), <https://doi.org/10.5281/zenodo.21767033>.
- [2] H. H. Hoos and T. Stützle, “SATLIB: An Online Resource for Research on SAT,” in *SAT 2000*, IOS Press (2000), <https://www.cs.ubc.ca/~hoos/SATLIB/>.
- [3] J. N. Hooker and V. Vinay, “Branching Rules for Satisfiability,” *Journal of Automated Reasoning* **15**, 359–383 (1995).
- [4] R. G. Jeroslow and J. Wang, “Solving Propositional Satisfiability Problems,” *Annals of Mathematics and Artificial Intelligence* **1**, 167–187 (1990).
- [5] M. W. Moskewicz *et al.*, “Chaff: Engineering an Efficient SAT Solver,” in *DAC 2001*, 530–535 (2001).